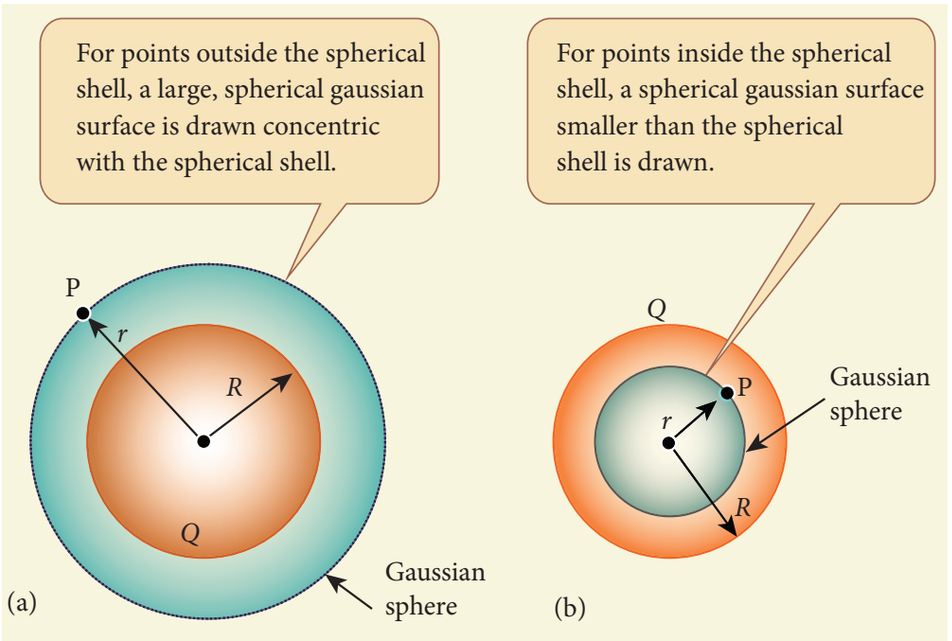


Figure 1.40 The electric field due to a charged spherical shell

the total charge enclosed by this Gaussian surface is Q. Applying Gauss law

$$\oint_{\text{Gaussian surface}} \vec{E}\cdot d\vec{A} = \frac{Q}{\epsilon_0} \qquad (1.73)$$

The electric field $\vec{E}$ and $d\vec{A}$ point in the same direction (outward normal) at all the points on the Gaussian surface. The magnitude of $\vec{E}$ is also the same at all points due to the spherical symmetry of the charge distribution.

$$\text{Hence } E \oint_{\text{Gaussian surface}} dA = \frac{Q}{\epsilon_0} \qquad (1.74)$$

But $\oint_{\text{Gaussian surface}} dA =$ total area of Gaussian surface $= 4\pi r^2$. Substituting this value in equation (1.74)

$$E\cdot 4\pi r^2 = \frac{Q}{\epsilon_0}$$

$$E\cdot 4\pi r^2 = \frac{Q}{\epsilon_0} \quad \text{(or)} \quad E = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}$$

In vector form,

$$\vec{E} = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}\hat{r} \qquad (1.75)$$

The electric field is radially outward if $Q > 0$ and radially inward if $Q < 0$. From equation (1.75), we infer that the electric field at a point outside the shell will be the same as if the entire charge Q is concentrated at the centre of the spherical shell. (A similar result is observed in gravitation, for gravitational force due to a spherical shell with mass M)

Case (b): At a point on the surface of the spherical shell ($r = R$)

The electrical field at points on the spherical shell ($r = R$) is given by

$$\vec{E} = \frac{Q}{4\pi\epsilon_0 R^2}\hat{r} \qquad (1.76)$$

Case (c): At a point inside the spherical shell ($r < R$)

Consider a point P inside the shell at a distance r from the centre. A Gaussian

12th Physics_Vol-1_Unit 1_English Medium.indd 45 08-12-2022 11:03:48